

Week 3

- Previous Bonus Problem
- Proof Patterns

Bonus Week 2

Prove that the formula $F = ((A \rightarrow B) \rightarrow A) \rightarrow A$ is a tautology, i.e. that $\models F$, using a sequence of **at most 9** equivalence transformations.

$$\begin{aligned} F &= ((A \rightarrow B) \rightarrow A) \rightarrow A \\ &\equiv ((\neg A \vee B) \rightarrow A) \rightarrow A && (F \rightarrow G \equiv \neg F \vee G) \\ &\equiv (\neg(\neg A \vee B) \vee A) \rightarrow A && (F \rightarrow G \equiv \neg F \vee G) \\ &\equiv ((\neg\neg A \wedge \neg B) \vee A) \rightarrow A && (\text{de Morgan's rule}) \\ &\equiv ((A \wedge \neg B) \vee A) \rightarrow A && (\text{double negation}) \\ &\equiv (A \vee (A \wedge \neg B)) \rightarrow A && (\text{commutativity of } \vee) \\ &\equiv A \rightarrow A && (\text{absorption}) \\ &\equiv \neg A \vee A && (F \rightarrow G \equiv \neg F \vee G) \\ &\equiv A \vee \neg A && (\text{commutativity of } \vee) \\ &\equiv \top && (F \vee \neg F \equiv \top) \end{aligned}$$

Move on predicate logic: Interpretation

Formula + Interpretation $\rightarrow$ Statement

- Without Interpretation: truth value can vary
- With Interpretation: fixed truth value: T/F

From last week:

$$U = \mathbb{N}$$

$$\forall x (x > 3) \quad \text{statement. not formula}$$

$$\forall x (x > 3) \wedge (y > 3) \quad \text{formula. truth value depends on } y$$

$$y = 0$$

$$\forall x \in \mathbb{N} (x > 3) \quad \text{Instead define the universe}$$

Extra:

Definition 6.32. Every occurrence of a variable in a formula is either *bound* or *free*. If a variable x occurs in a (sub-)formula of the form $\forall x G$ or $\exists x G$, then it is bound, otherwise it is free.⁵² A formula is *closed*⁵³ if it contains no free variables.

Definition 6.35. A interpretation (structure) $\mathcal{A}$ is *suitable* for a formula F if it defines all function symbols, predicate symbols, and freely occurring variables of F .

Proof Patterns

- Direct proof

$$S \Rightarrow T$$

- Indirect proof (Contraposition)

Definition 2.14. An indirect proof of an implication $S \Rightarrow T$ proceeds by assuming that T is false and proving that S is false, under this assumption.

$$A \rightarrow B \equiv \neg B \rightarrow \neg A$$

Example: if n is odd, then n^2 is odd.

Assume n^2 is even $\Rightarrow \dots \Rightarrow n$ is even

- Modus Ponens

Definition 2.15. A proof of a statement S by use of the so-called *modus ponens* proceeds in three steps:

1. Find a suitable mathematical statement R .
2. Prove R .
3. Prove $R \Rightarrow S$.

$$A \wedge (A \rightarrow B) \models B$$

- Case Distinction

Definition 2.16. A proof of a statement S by *case distinction* proceeds in three steps:

1. Find a finite list $R_1, \dots, R_k$ of mathematical statements (the *cases*).
2. Prove that at least one of the R_i is true (at least one case occurs).
3. Prove $R_i \Rightarrow S$ for $i = 1, \dots, k$.

Lemma 2.8. For every k we have

$$(A_1 \vee \dots \vee A_k) \wedge (A_1 \rightarrow B) \wedge \dots \wedge (A_k \rightarrow B) \models B.$$

Exercise: Let $p > 3$ be a prime number. Prove that

$$p^2 - 1$$

is divisible by 24.

$p \div 6$ remainder: ~~0~~ 1 ~~2~~ ~~3~~ ~~4~~ 5

Case 1: $p = 6k + 1 \quad k \in \mathbb{N}$

$$1^2 - 1 = \underline{0}$$

~~$p = 2k$~~ X
 ~~$p = 2k + 1$~~

Case 2: $p = 6k + 5 \quad k \in \mathbb{N}$ $24 \nmid 4k^2 + 4k$

$$5^2 - 1 = \underline{24}$$

$$\frac{p^2 - 1}{24} = \frac{36k^2 + 12k}{24} = \frac{3k^2 + k}{2} \in \mathbb{Z} \quad \begin{array}{l} k: \text{even} \\ k: \text{odd} \end{array}$$

$$\frac{p^2 - 1}{24} = \frac{36k^2 + 60k}{24} = \frac{3k^2 + 5k}{2} \in \mathbb{Z} \quad \begin{array}{l} k: \text{even} \\ k: \text{odd} \end{array}$$

Example: Exam HS14

(*) Show that for every n in $\mathbb{N}$, $n^3 + 2n$ is divisible by 3.

Cases:

$$n = (3k+1)^3 = \text{---} + 1^3 \rightarrow 1^3 + 2 \cdot 1 = 3$$

$$(3k+2)^3 = \text{---} + 2^3 \rightarrow 2^3 + 2 \cdot 2 = 12$$

• Existence Proof

Prove there are infinitely many primes

Assume there are finitely many primes $p_1, p_2, \dots, p_m$ (proof by contradiction)

Construct a prime p greater than all of them. (existence proof)

new prime

$$P_i (P_1 P_2 \dots P_m) + 1 \neq P_i \cdot k, k \in \mathbb{N}$$

$$P = P_1 P_2 \dots P_m + 1$$

$$P \neq P_i \cdot k, k \in \mathbb{N} \text{ for all } i$$

$$24 = 2 \cdot 2 \cdot 2 \cdot 3$$

$$25 = 5 \cdot 5$$

• Counterexample

Example 2.34. Prove or disprove that for every integer n , the number $n^2 - n + 41$ is prime, i.e., prove

$$\forall n \text{ prime}(n^2 - n + 41).$$

$$n = 41$$

$$41^2 - 41 + 41$$

- Induction true for all $n \in \mathbb{N}$ $S(n)$
 - Base case $n=0$ $S(0)$
 - Induction step $n=k \rightarrow n=k+1$ $S(k) \Rightarrow S(k+1)$
 - Example: Exam HS14

(*) Prove that $n! > 2^n$ for all $n \geq 4$. $n! = n \cdot (n-1) \cdot \dots \cdot 2 \cdot 1$

Base case: $4! > 2^4$ $24 > 16$ ✓

Induction hypothesis: Assume the statement holds for some $n \geq 4$, i.e. $n! > 2^n$

Induction step: $(n+1)! = (n+1)n!$
 $> (n+1)2^n$ (IH)
 $> 2 \cdot 2^n$ $n \geq 4 \Rightarrow n+1 > 2$
 $= 2^{n+1}$

- Soundness of proof patterns (Exercise 3.7)

Proof by contradiction:

Definition 2.17. A proof by contradiction of a statement S proceeds in three steps:

1. Find a suitable mathematical statement T .
2. Prove that T is false.
3. Assume that S is false and prove (from this assumption) that T is true (a contradiction).

Step 1: Write the statement you want to prove as formula G

$$S \triangleq A, \quad G = A$$

Step 2: Write the proof steps as formula F

$$T \triangleq B, \quad F = \neg A \rightarrow B$$

Step 3: Prove $F \models G$

$$\neg A \rightarrow B \models A \quad \text{Check the truth table}$$

Useful cases

- $\forall x S(x)$
 - Induction, when $u \in \mathbb{N}$
 - Contradiction, $\exists x (\neg S(x))$
- $\neg \forall x S(x)$
 - Counterexample
- $\exists x S(x)$
 - Existence proof